Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/Alan/Static · 35 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/Alex/Static · 35 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/Georgia/Static · 35 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/Henry/Static · 35 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/Jestin/Static · 36 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/Joselyn/Static · 30 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/Josh/Static · 35 trials
Training data path 8	/home/jestin/ThesisRepo/ML/NewTestData/Nat/Static · 37 trials
Total training paths	8 · 278 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/Marcus/Static
Report output dir	/home/jestin/ThesisRepo/ML/saved_reports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	Flex
Sensor columns	20 channels
Classes	7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 5.0 Hz · order 4 · fs 30.0 Hz
Normalisation	standard
Feature mode	stats → feature dim 160
Test size · seed · CV folds	0.2 · random_state=42 · 10-fold stratified
Sample counts	train (post-aug)=444 · test=56

Augmentation (training only)

Enabled	yes
Copies per train trial	1
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.8, 1.2)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (10-fold)	CV std	Hold-out test acc
Random Forest	0.6106	0.0526	0.8929
Stacking (SVM+RF+KNN+LogReg)	0.8628	0.0660	0.8393
LDA	0.9662	0.0381	0.7500
Voting Soft (SVM+RF+KNN+LogReg)	0.8786	0.0614	0.5357
Logistic Regression	0.9124	0.0583	0.4821
SVM (RBF)	0.7706	0.0646	0.3571
SVM (Linear)	0.9327	0.0570	0.3393
KNN	0.3601	0.0722	0.2857

Per-class hold-out F1

Class	Random Forest	Stacking	LDA	Voting (soft)	Logistic Regression	SVM (RBF)	SVM (Linear)	KNN	Support
Double_Closed_Fist	1.000	1.000	0.625	0.750	0.615	0.588	0.444	0.276	9
Double_Nothing	0.923	0.714	0.833	0.533	0.444	0.500	0.364	0.500	7
Double_Okay	0.875	0.750	0.545	0.636	0.636	0.526	0.583	0.267	8
Double_Open_Palm	0.875	0.824	0.714	0.462	0.500	0.000	0.182	0.200	8
Double_Phone	0.889	0.889	0.875	0.700	0.625	0.333	0.375	0.381	8
Double_Pistol	0.889	0.800	0.875	0.000	0.133	0.000	0.000	0.133	8
Double_Spiderman	0.769	0.857	0.875	0.400	0.375	0.250	0.000	0.200	8
macro avg	**0.889**	**0.833**	**0.763**	**0.497**	**0.476**	**0.314**	**0.278**	**0.280**	**56**

External test (NewTestData)

Algorithm	External test acc
Random Forest	0.7143
Stacking (SVM+RF+KNN+LogReg)	0.6857

LDA	0.6000
Voting Soft (SVM+RF+KNN+LogReg)	0.4857
Logistic Regression	0.4286
SVM (Linear)	0.4000
SVM (RBF)	0.2857
KNN	0.0857

Per-class external accuracy

Class	Random Forest	Stacking	LDA	Voting (soft)	Logistic Regression	SVM (Linear)	SVM (RBF)	KNN	Support
Double_Closed_Fist	1.000	1.000	0.800	0.400	0.400	0.200	0.000	0.200	5
Double_Nothing	0.600	0.400	0.400	0.600	0.400	0.800	0.800	0.200	5
Double_Okay	1.000	1.000	0.400	0.600	0.400	0.600	0.200	0.000	5
Double_Open_Palm	0.000	0.000	0.400	0.200	0.200	0.200	0.200	0.000	5
Double_Phone	1.000	0.800	0.800	0.200	0.400	0.200	0.000	0.000	5
Double_Pistol	1.000	1.000	1.000	0.800	0.800	0.600	0.800	0.200	5
Double_Spiderman	0.400	0.600	0.400	0.600	0.400	0.200	0.000	0.000	5